Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… ☐ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. ☐ Read or plot how pH links to rate. Say only what the data supports. ☐ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Aspire

1. Time and rate

A trial takes twice as long to reach the same endpoint. What happens to its reciprocal relative rate?

Compare 1/t with 1/(2t). Starting amount and endpoint must be matched.

2. Read the endpoint

Why is 1/90 s^{-1} an endpoint-based estimate rather than a direct instantaneous rate measurement?

The endpoint records one sampled stage of change; rate may vary during a reaction.

3. Show the working

For 120 s, compare 1/t with 1000/t. May the two values share one unlabelled graph axis?

The two conventions differ by a scale factor of 1000. This lesson uses 1/t only.

4. Match the instruction

A graph has one result at each pH. Name one extra thing an EVALUATE answer needs beyond describing its peak.

Evaluate = use a strength or limitation to reach a supported judgement.

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Exit Ticket · Aspire

Evidence

Illustrative matched-condition data: pH 6 endpoint 120 s; pH 7 endpoint 90 s. A separate pH 7 repeat gave 120 s. Relative rate = 1 ÷ time in seconds (s^{-1}). Samples were 30 s apart; pH 4–8 were tested.

1. Does the pH 7 repeat strengthen the claim of an exact optimum? Use a rate and a comparison.

Its 120 s result is a separate observation at the same pH. Compare with pH 6.

2. Give one change to the method that would help with a specific problem. Say why this graph still cannot prove permanent enzyme denaturation.

Better endpoint evidence is one thing. Direct tests for a change in the molecules, or for recovery, are another. Show how these differ.

Under the heading Exit ticket: answer the short questions in your book.